Review Questions for Lecture 15

Set up an appropriate box model for questions 1-10 and answer the following questions:

- How many tickets are in the box?
- What are the numbers on the tickets?
- What is the average of the box?
- What is the SD of the box?
- How many tickets are drawn?
- Are the tickets drawn with or without replacement?

1. You repeat the following 8 times: You roll a die and flip a coin. If the coin lands heads, you win the amount on the die. If the coin lands tails, you lose 4 dollars. You want to track your total winnings.

2. You roll a die 8 times and want to track the number of fives.

3. There are 100 people in a room. Each person flips a coin. Anyone who gets tails leaves the room. You want to track the number of people remaining in the room.

4. There are 100 people in a room. Each person flips a coin. Anyone who gets tails leaves the room. Then, of the people who remain, each person rolls a die. Anyone who gets a three leaves the room. You want to track the number of people remaining in the room.

5. Each day a factory produces 10,000 red M&Ms, 10,000 green M&Ms, 10,000 brown M&Ms, and 10,000 blue M&Ms. The M&Ms are all mixed together in a big vat. Then 30 M&Ms are taken from the vat and put into a bag. You want to track the number of green M&Ms in a bag?

6. You repeat the following 16 times: You roll a die and flip a coin. If the coin lands heads and the number on the die is odd, you win the amount on the die. If the coin lands tails, you lose 4 dollars. You want to track your net winnings?

7. In your pocket you have 3 pennies, 2 nickels, 4 dimes and a quarter. You reach into your pocket and randomly draw out 4 coins. You want to track the total amount of money that you pull out. Given that these coins are different sizes, the idea that the four coins you pull out would be random isn't realistic.

8. For a particular basketball player, the probability of making a free throw is 7 out of 10. The basketball player will take two free throws. You want to track the number of points the player scores. Each successful free throw gets one point.

Note that the R command `sample(data,size,replace=FALSE)` outputs a random sample without replacement of size `size` from the vector `data`. You may find this command useful in some of the exercises below.

9. You repeat the following 16 times: You roll a die and flip a coin. If the coin lands heads and the number on the die is odd, you win the amount on the die. If the coin lands tails, you lose 4 dollars. What is the expected value and standard error for your winnings?

10. In your pocket you have 3 pennies, 2 nickels, 4 dimes and a quarter. You reach into your pocket and randomly draw out 4 coins.
    a. What is the expected value for the total amount of money you draw from your pocket?
    b. Can you use the formulas from class to calculate the standard error for the total amount of money from your pocket? Why or why not?
    c. Use either the formulas from class or simulation to estimate the standard error for the total amount of money from your pocket. Include your code and output in your submission if you use simulation to estimate the standard error.

11. For a particular basketball player, the probability of making a free throw is 7 out of 10. The basketball player will take two free throws. What is the expected value and standard error for the number of points that the player will score?

12. Suppose that the basketball player in problem 8 takes 100 free throws.
    a. How does the box model change from problem 8?
    b. What is the expected value and standard error for the total number of points that the player scores?
    c. What is the expected value and standard error for the percentage of free throws that the player makes?
    d. Calculate the probability that the player would make 65 or more free throws?
    e. Estimate the probability in d. assuming that the percentage of free throws made is normally distributed.
    f. Is the value in e. close to the value in d.?

13. A group of 20 people play Wheel of Fortune. See Review_questions9-10.pdf problem 1. Each person at the party plays three times; in between the times they re-spin the wheel.
    a. What would be the expected value for the percentage of people who win $1,000,000?
    b. Can you use the formulas from class to calculate the standard error for the percentage of people who win $1,000,000? Why or why not?
    c. Use either the formulas from class or simulation to estimate the standard error for the percentage of people who win $1,000,000. Include your code and output in your submission if you use simulation to estimate the standard error.

14. Suppose that in problem 13, instead of respinning, the player just moves the wheel forward one spot each time.

a. What would be the expected value for the percentage of people who win $1,000,000?

b. Can you use the formulas from class to calculate the standard error for the percentage of people who win $1,000,000? Why or why not?

c. Use either the formulas from class or simulation to estimate the standard error for the percentage of people who win $1,000,000. Include your code and output in your submission if you use simulation to estimate the standard error.

15. There is a rule of thumb that states that the percentage p of 1's in n draws with replacement from a Zero-One box is approximately normally distributed when $n*p>5$ and $n*(1-p)>5$. This problem investigates this rule of thumb with a couple of examples.

a. Suppose that you take 4 draws with replacement from a box with 90% red balls and 10% green balls. What is the chance of getting no green balls?

b. What would be your estimate of the chance in part a. if you assumed that the percentage of green balls is approximately normally distributed? Remember to use the continuity correction, so that you'll find the chance of 0.5 or fewer green balls.

c. Does this example follow the rule of thumb? Is the estimate in part b. close to the exact value in part a.?

d. Repeat parts a)-c) with 400 draws and the chance that the number of green balls is at least 30 and at most 35. Remember to use the continuity correction when you are using the normal approximation. You'll find the chance of 29.5 to 35.5 green balls with the correction.

16. Derive the short-cut formula on page 298 of the text.